

CFG Construction

1) Any no. of a's.

$$L = \{ \epsilon, a, aa, \dots \}$$

$$S \rightarrow aS$$

$$S \rightarrow \epsilon$$

2) Any no. of a's b's.

$$L = \{ \epsilon, a, b, aa, ab, ba, \dots \}$$

$$S \rightarrow aS \mid bS \mid \epsilon$$

3) At least 2 a's. $= R = (a+b)^* a (a+b)^* a (a+b)^*$

$$L = \{ aa, aba, abba, baab, \dots \}$$

$$S \rightarrow AaAaA$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

4) One occurrence of ooo

$$S \rightarrow AoooA$$

$$A \rightarrow oA \mid \epsilon$$

5) $a^n b^n \quad n \geq 1$

$$L = \{ ab, aabb, \dots \}$$

$$S \rightarrow aSb$$

$$S \rightarrow ab$$

6) $a^n b^{2n} \quad n \geq 1$

$$S \rightarrow aSbb$$

$$S \rightarrow abb$$

7. $L = \{ w c w^T \}$

$\Rightarrow$ odd length
palindrome.

$$S \rightarrow aSa$$

$$S \rightarrow bSb$$

$$S \rightarrow c$$

$$asa \mid bsb \mid a \mid b \mid \epsilon$$

8) $a^n b^m c^n \quad n, m \geq 1$

$$S \rightarrow aSc \mid aAc$$

$$A \rightarrow bAb$$

$$\begin{array}{c} S \\ \swarrow \searrow \\ aSc \\ \swarrow \searrow \\ aSc \end{array}$$

9) $L = \{ a^n b^m c^m \mid n, m \geq 1 \}$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid aAb$$

$$B \rightarrow c \mid Bc$$

9) $L = \{ a^n b^m c^m d^n \mid n \geq 1, m \geq 1 \}$

$$S \rightarrow aSd \mid aAd$$

$$A \rightarrow bAc \mid bc$$

10) $L = \{ a^n b^m c^n d^m \mid n, m \geq 1 \}$

11) $L = \{ a^n b^n c^m d^m \mid n, m \geq 1 \}$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow cBd \mid cd$$

12) $L = \{ a^{m+n} b^m c^n \mid n, m \geq 1 \}$

$$S \rightarrow aSb \mid aSb$$

$$S \rightarrow aSb \mid aSb$$

$$A \rightarrow aAb \mid ab$$

$$L = \{0^n 1^m 2^{n+m} \mid n, m \geq 1\}$$

$$= \{0^n 1^m 2^n 2^m\}$$

$$= \{0^n 1^m 2^m 2^n\}$$

$$S \rightarrow 0A2 \mid 0S2$$

$$A \rightarrow 12 \mid 1A2$$

$$L = \{a^n b^n \mid n \geq 1\}$$

$$S \rightarrow abc \mid A$$

$$A \rightarrow aABc \mid abc$$

$$S \rightarrow AB \mid \epsilon$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow bBc \mid bc$$

$$L = \{a^{2n} b^n \mid n \geq 0\}$$

$$S \rightarrow aasb \mid \epsilon$$

$$L = \{a^{2n} b^n \mid n \geq 1\}$$

$$S \rightarrow aasb \mid aab$$

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

$$aSbb \mid abb$$

$$L = \{a^n b^n\}$$

$$S \rightarrow aAb \mid ab$$

$$a^n b^m c^m$$

$$a^n b^n b^m c^m$$

$$n=2 \quad m=3$$

$$a^2 b^5 c^3$$

$$aabb bbb ccc$$

$$a^n b^m b^n c^m$$

$$a^n b^n b^m c^m$$